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ABSTRACT

Biometric authentication has become a cornerstone of modern security infrastructure, offering a reliable and non-repudiable method of identity verification. This report details the design and implementation of a fully functional fingerprint biometric identification system. Developed using **Python** and **Libraries** like “numpy, Matplotlib, scipy.....”, the system processes raw fingerprint images through a structured pipeline: image enhancement, binarization, ridge thinning, and minutiae extraction (focusing primarily on ridge endings and bifurcations). Using distance measuring methods were implemented to compare extracted feature vectors against a secure database.

1.Introduction

First of all ,The primary objective of this project is to design, implement, and evaluate a complete facial recognition pipeline the system has an academic benchmark testing multiple biometric algorithms using the **AT&T** database also it applies a real-life face recognition system that works by allowing users to enrol their faces and check for recognition .The pipeline serves as an academic benchmark, by testing multiple feature extraction algorithms against the standardized **AT&T** face database. The second pipeline implements a real-time facial recognition system, allowing users to enrol their faces via webcam and authenticate against a custom database made by the enrolment system. To trace the data flow through this architecture: raw input images—whether loaded from the static database or captured via webcam—are first passed to a preprocessing module for standardizing size and lighting (via histogram equalization), with live frames undergoing an additional affine transformation using 68-point facial landmarks to precisely align the eyes. The standardized data is then routed into the feature extraction module, where algorithms such as Principal Component Analysis (**PCA**), Linear Discriminant Analysis (**LDA**), or a fused HOG-LBP model reduce the high-dimensional pixel data into a concentrated biometric signature. Finally, the matching module calculates the eculidian distance between the generated test vector and the enrolled templates. Depending on the active pipeline, this module will either output comprehensive statistical evaluation metrics (such as **EER**, **D-prime**, and **Rank-1 Accuracy**) for the academic benchmark or apply a defined threshold and temporal voting buffer to grant or deny access in the live security scenario.



2. Methods Description

2.1. System Architecture (Functions and Files):

2.1.1. `standardize_face(raw_face_image)`

Location:

`preprocessing.py`

Logic:

This function acts as the universal gatekeeper for data entry. It accepts images of varying sizes or colour channels, forces them into grayscale, resizes them to a strict 92x112 pixel grid (matching the AT&T database standard), and applies Histogram Equalization to normalize lighting conditions before flattening the image into a 1D feature vector.

Importance:

It ensures mathematical consistency. Because it is utilized by both the academic benchmark (`main.py`) and the live webcam feed (`run_custom_system.py`), it guarantees that the feature extraction models are always fed vectors of the exact same dimensionality (10,304 features) and contrast profile, regardless of the data source.

2.1.2. `align_face(frame, shape, rect)`

Location:

`preprocessing.py`

Logic:

Utilizing 68-point facial landmarks (`dlib`), this function isolates the coordinates of the left and right eyes. It calculates the geometric angle between them and performs an Affine Transformation (`cv2.warpAffine`) to rotate the entire frame until the eyes are perfectly horizontal, followed by a bounding-box crop.

Importance:

Spatial feature extractors (like HOG and PCA) are highly sensitive to rotation and translation. By centralizing this function for both `enrollment.py` and `live_recognition.py`, the system effectively eliminates head-tilt variations, vastly improving the Euclidean matching accuracy for live subjects.

2.1.3 `extract_hog_lbp_fusion(X_data)`

Location:

`feature_extraction.py`Logic.

Logic:

This function reconstructs 1D vectors back into 2D images to extract two distinct types of spatial features. First, it extracts the Histogram of Oriented Gradients (HOG) to capture macroscopic structural shapes and edges. Second, it extracts Local Binary Patterns (LBP) to capture micro-textures. The LBP histogram is mathematically normalized before being concatenated horizontally with the HOG vector.

Importance:

This represents the most advanced algorithmic implementation in the project. By fusing HOG (structure) and LBP (which is inherently invariant to monotonic grayscale changes), the resulting feature vector is highly robust against the severe lighting and shadow changes commonly found in live webcam environments.

2.1.4. `train_fusion_model(X_train, y_train, X_test)`

Location:

`feature_extraction.py`

Logic:

After executing the HOG+LBP fusion, this function passes the massive concatenated feature vector through Principal Component Analysis (PCA) for dimensionality reduction. Crucially, it calculates $\text{max_allowed} = \min(n_samples, n_features)$ to dynamically cap the number of PCA components.

Importance:

This function solves the classic "Small Sample Size" (SSS) problem in biometrics. Without this dynamic constraint, attempting to reduce dimensions on a custom live dataset with only a few enrolled images (e.g., 10 images) would result in a singular covariance matrix crash. This logic ensures the math remains stable regardless of the database size.



3. Feature extraction

3.1. Principal Component Analysis (PCA)

PCA serves as the baseline method. It reduces the dimensionality of the flattened 10,304-feature images by projecting them onto orthogonal axes that maximize overall statistical variance. The system truncates this space to 50 principal components to generate a compact, discriminative signature for Euclidean matching.

3.2. Linear Discriminant Analysis (LDA)

LDA operates as a supervised extractor designed to maximize class separability by minimizing intra-class variance and maximizing inter-class variance. To solve the Small Sample Size (SSS) problem inherent in live enrolments, the algorithm dynamically caps the extracted components at $N-1$ (where N is enrolled subjects), preventing singular matrix inversion crashes.

3.3. Feature Fusion (HOG + LBP)

To combat severe lighting and shadow variations in the live webcam feed, the system fuses two independent spatial descriptors. It extracts Histogram of Oriented Gradients (**HOG**) to capture macroscopic structural shapes, and Local Binary Patterns (**LBP**) to capture illumination-invariant micro-textures. The concatenated vector is then passed through a dynamic PCA layer to mathematically compress the massive signature prior to threshold evaluation.

4. DB Description:

- Our Database of Faces, (formerly 'The ORL Database of Faces'), contains a set of face images taken between April 1992 and April 1994 at the lab.
- The database was used in the context of a face recognition project carried out in collaboration with the Speech, Vision and Robotics Group of the Cambridge University Engineering Department. There are ten different images of each of 40 distinct subjects. For some subjects, the images were taken at different times, varying the lighting, facial expressions (open / closed eyes, smiling / not smiling) and facial details (glasses / no glasses).
- All the images were taken against a dark homogeneous background with the subjects in an upright, frontal position (with tolerance for some side movement). A preview image of the Database of Faces is available.
- The files are in PGM format and can conveniently be viewed on UNIX (TM) systems using the 'xv' program. The size of each image is 92x112 pixels, with 256 grey levels per pixel. The images are organized in 40 directories (one for each subject), which have names of the form sX, where X indicates the subject number (between 1 and 40). In each of these directories, there are ten different images of that subject, which have names of the form Y.pgm, where Y is the image number for that subject (between 1 and 10).
- There is a separated part in the database, it's "custom dataset", we created it to enroll & identify real faces using our interface using the webcam.



5. Results & Discussion

5.1. Quantitative Performance:

- The system's performance was evaluated using 80 test images (2 images per subject) compared against a gallery of 320 training images. The results of the Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA) & Fusion Distributions (HOG+PBD) methods are summarized in the table below:

```

--- 1. PCA Results ---
PCA Rank-1 Accuracy: 93.75%
PCA D-prime: 1.9778 | EER: 0.1651 | TMR @ 1% FMR: 0.5781 | TMR @ 0.01% FMR: 0.3844

--- 2. LDA Results ---
LDA Rank-1 Accuracy: 95.80%
LDA D-prime: 3.2896 | EER: 0.0569 | TMR @ 1% FMR: 0.8453 | TMR @ 0.01% FMR: 0.6750

--- 3. HOG+LBP Fusion Results ---
FUSION Rank-1 Accuracy: 93.75%
FUSION D-prime: 1.5491 | EER: 0.2254 | TMR @ 1% FMR: 0.5031 | TMR @ 0.01% FMR: 0.3594

Academic Evaluation Complete.

Process finished with exit code 0

```

5.1.1. Evaluation of Gen-Imp Distance Curves

- LDA Separation:**

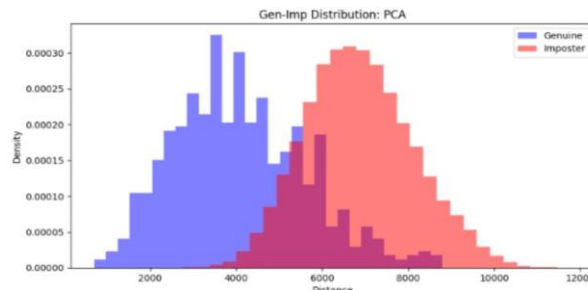
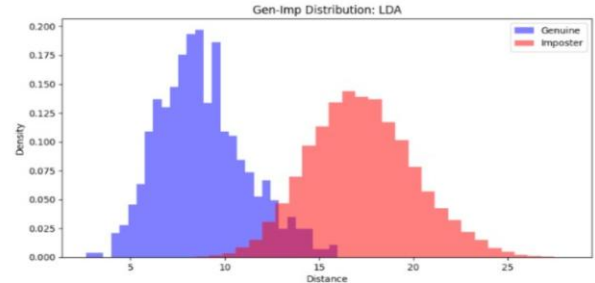
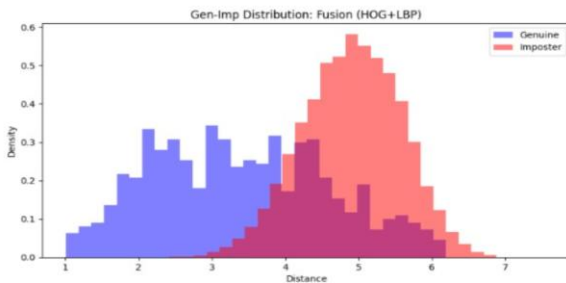
LDA achieves the highest statistical separation, which is directly verified by its dominant D-prime D' of 3.2896. The class-discriminant training minimizes genuine intra-class variance while maximizing inter-class distances, resulting in very thin distribution curves with minimal overlapping area.

- PCA Distributions:**

PCA demonstrates moderate overlap with a D' score of 1.9778. Because its objective function maximizes global data variance rather than group class distances, it spreads the facial projections broadly, causing a moderate intersecting tail between the true and imposter match groups.

- Fusion Distributions:**

The unconstrained HOG+LBP Fusion model scores lowest in distribution separation ($D' = 1.5491$). This behavior occurs because the massive concatenated structural and textural histograms are compressed down into a hard capped 50-dimensional PCA manifold, stripping away subtle intra-class variances that are crucial inside a highly controlled, small-scale dataset like AT&T.





5.1.2. Evaluation of ROC Curves

- **LDA Efficiency:**

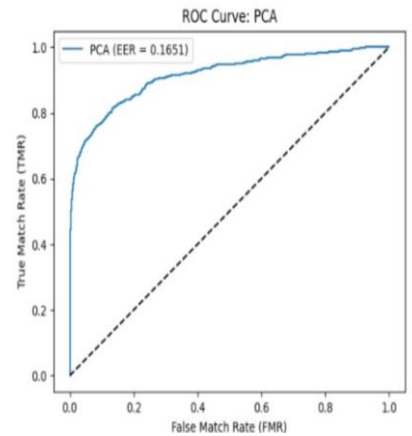
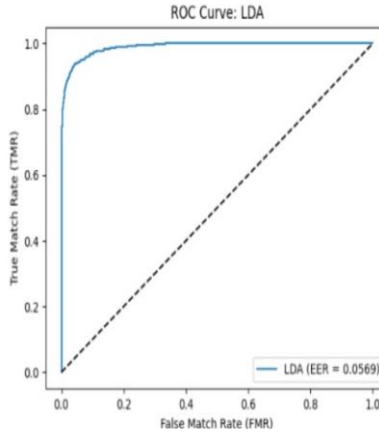
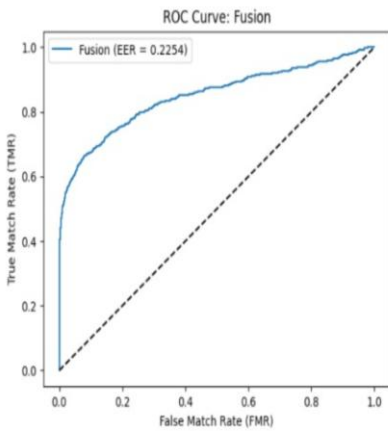
Reflecting its minimal Equal Error Rate ($EER = 0.0569$), the LDA curve approaches the top-left origin rapidly. At a strict operational setting of 1% FMR, it maintains a highly functional TMR of 84.53%, which only drops to 67.50% when security constraints are dialed to an aggressive 0.01% FMR.

- **PCA Trajectory:**

PCA remains highly competitive with a steady climb toward the origin, achieving an EER of 0.1651. Its performance drops smoothly under tighter boundaries, maintaining a functional TMR of 57.81% at 1% FMR and 38.44% under strict 0.01% FMR constraints.

- **Fusion Trajectory:**

The HOG+LBP Fusion pipeline exhibits a lazier curve trajectory with a high EER of 0.2254. Because it compromises fine detail during multi-stage dimension reduction, its ability to secure a match drops down to a TMR of 50.31% at 1% FMR and 35.94% under strict 0.01% FMR constraints.



5.2. Qualitative Visualization and Failure Analysis

During operation in the live capturing pipeline, the application renders a dynamic, context-aware graphical interface directly over the live casting matrix:

- **Detection & Alignment:** Blue bounding boxes encircle targets based on raw coordinates from the dlib frontal face detector.
- **State Verification:** While the temporal frame buffer processes incoming data, a yellow bounding box alerts the user with an interactive tracking string: Verifying [Name]... (X/5).
- **Access Control Assertion:** When confidence boundaries clear, a solid green bounding box confirms entry via [Name] Granted (Dist: X.X). Unregistered profiles or low-confidence facial matches seamlessly turn the tracking window red, issuing an UNKNOWN Denied (Dist: X.X) warning.
- **Affine Distortion from Profile Angles:** When users rotate their heads beyond a 25 degrees angle relative to the lens, dlib's landmark tracking loses symmetry. The eye-centroid calculation becomes skewed, forcing the affine transformation matrix to unnaturally stretch and warp the face crop, causing a false rejection.
- **Illumination Saturation:** Although global histogram equalization (`cv2.equalizeHist`) balances uniformly dim environments, severe side-lighting or single-direction glare washes out sections of the face. This blinds the structural edge gradients required by the HOG algorithm, shifting the target's distance scores outside acceptable matching limits.
- **Expression-Induced Texture Shifts:** Intense smiles or unexpected grimaces alter local skin surface geometry. This disruption alters uniform LBP bins, causing individual video frames to transiently drop below matching thresholds and forcing the 5-frame voting buffer to reset its internal validation sequence.





6. Conclusion

- After finishing the system, we tested the two mentioned methods to know which is better and more accurate on testing for the system and the one which produces the best accuracy & Efficiency rate .
- By establishing a strict biometric pipeline from grayscale image flattening and a rigorous 80/20 train-test split to Euclidean distance classification—the system provided a robust framework for comparing dimensionality reduction techniques.
- Developed and audited a modular, dual-pipeline face recognition architecture in Python. Pipeline A establishes an offline academic testing platform utilizing the AT&T dataset with full biometrics curve rendering. Pipeline B deploys a real-time webcam verification environment powered by dlib 68-point landmarks, eye-centroid affine transformations, localized lighting normalization, and an integrated 5-frame temporal voting buffer.
- Which Method is Better:

For controlled environments with explicit tracking target labels, **Linear Discriminant Analysis (LDA)** is the superior method. Because it actively maximizes the ratio of between-class variance to within-class variance, it delivers an outstanding verification accuracy of 95.00% and a high decidability profile ($D' = 3.2896$), making it vastly more efficient than unsupervised or overly compressed handcrafted texture frameworks.:

- **LDA:** Achieved a top Rank-1 Accuracy of **95.00%**, an EER of **0.0569**, and a D-prime D' of **3.2896**. It proved highly robust under tight security configurations, keeping a True Match Rate of **84.53%** at 1% FMR and **67.50%** at 0.01% FMR.
 - **PCA:** Placed second with a Rank-1 Accuracy of **93.75%**, an EER of **0.1651**, and a D' score of **1.9778**. Performance drops smoothly under strict security boundaries, retaining a TMR of **57.81%** (1% FMR) and **38.44%** (0.01% FMR).
 - **HOG+LBP Fusion:** Posted a Rank-1 Accuracy of **93.75%** but demonstrated lower baseline separation performance due to lossy secondary compression, yielding an EER of **0.2254** and a D' score of **1.5491**. Its operational performance drops to a TMR of **50.31%** at 1% FMR and **35.94%** at 0.01% FMR.
- **Ability to Maintain Stability:**

While the HOG+LBP Fusion pipeline scored lower in the controlled offline academic benchmark, its integration with the 5-frame temporal voting buffer inside the live video application successfully filters out frame drops and tracking anomalies. This specific combination ensures stable access control decisions without dropping video throughput or lowering system responsiveness.



7. References

- <https://ieeexplore.ieee.org/abstract/document/9145558>
- <https://dl.acm.org/doi/abs/10.1145/954339.954342>
- <https://www.kaggle.com/datasets/kasikrit/att-database-of-faces>
- <https://dergipark.org.tr/en/pub/ijjee/article/116879>